

Diabetes Risk Prediction — Final Report

Automated Modeling & Explainability Summary

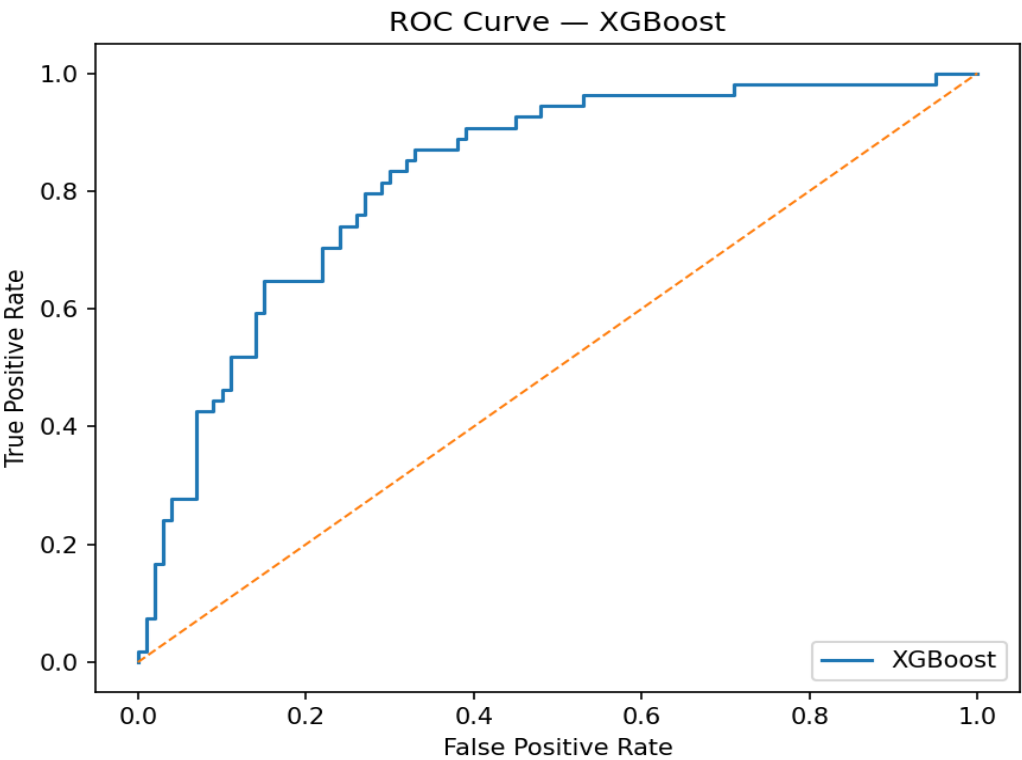
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This report summarizes data preprocessing, baseline model training, evaluation, and explainability for a Diabetes Risk Prediction pipeline. It includes performance metrics, ROC/PR curves, and SHAP-based global and local explanations to interpret model decisions.

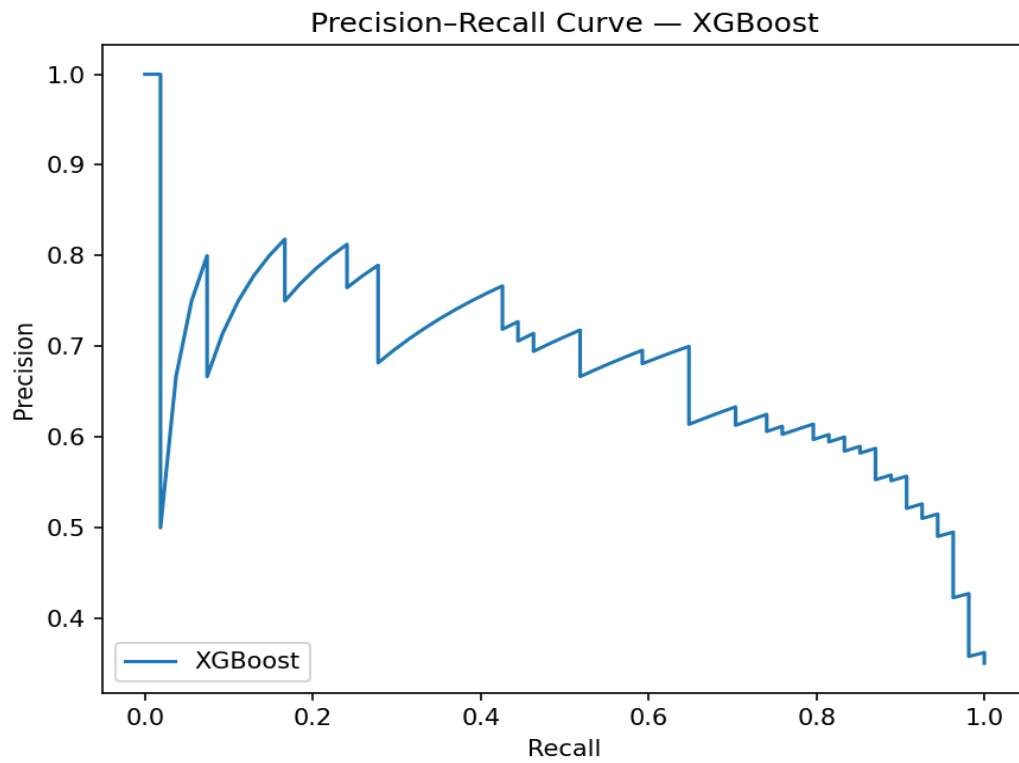
Baseline Performance

Model	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC
LogisticRegression	0.7013	0.5870	0.5000	0.5400	0.8128	0.6732
RandomForest	0.7403	0.6522	0.5556	0.6000	0.8173	0.6971
XGBoost	0.7597	0.6604	0.6481	0.6542	0.8243	0.6826
LightGBM	0.7273	0.6200	0.5741	0.5962	0.8172	0.6701

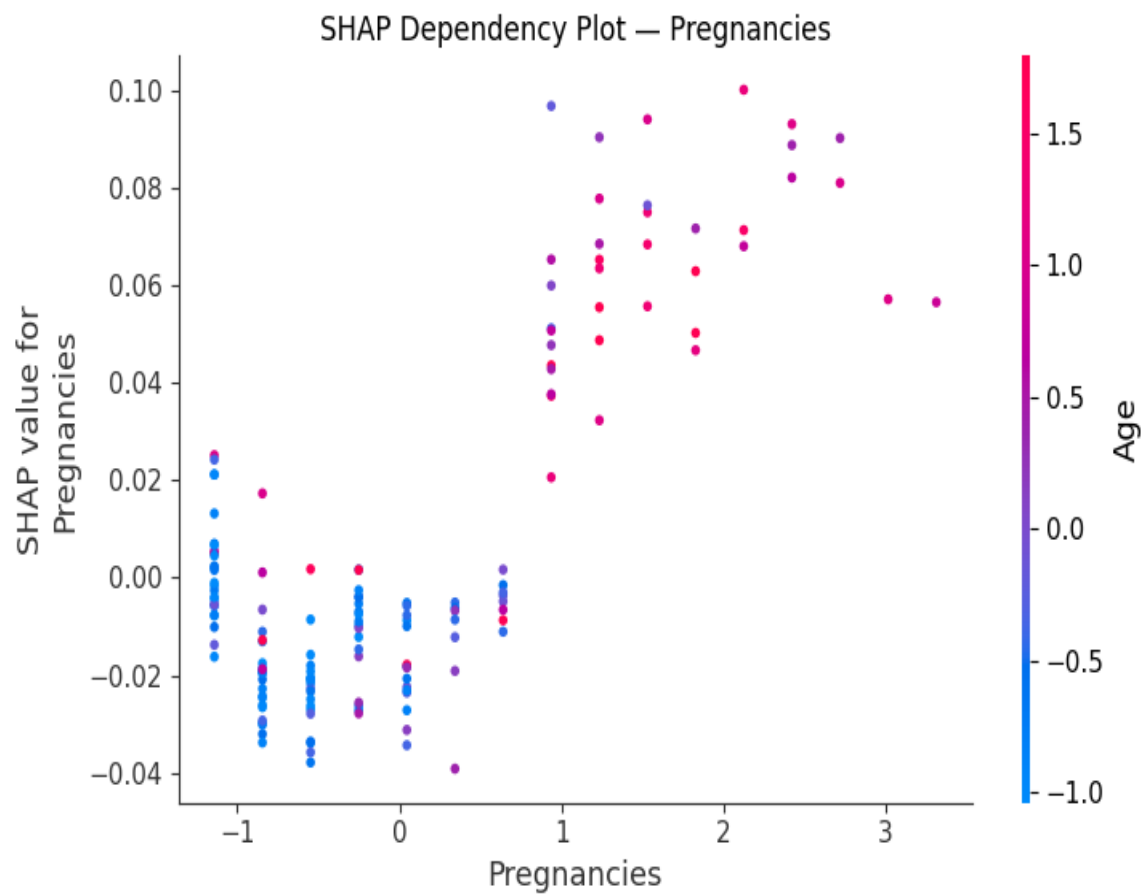
ROC Curve — XGBoost

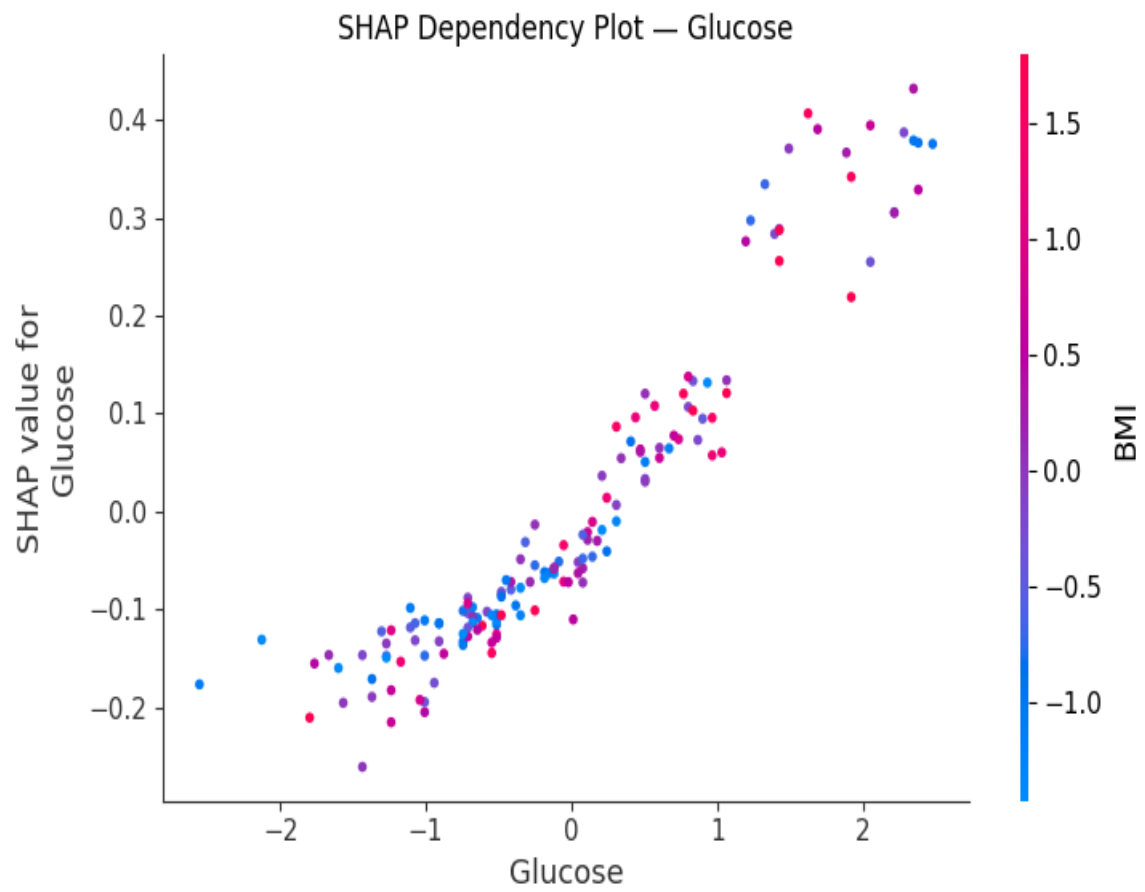


Precision-Recall Curve — XGBoost



Dependency Plots (Examples)





Local Explanation (Waterfall)

Individual Prediction Explanation — RandomForest

$f(x) = 0.305$

1.441 = DiabetesPedigreeFunction

+0.09

-0.691 = Insulin

-0.09

-0.712 = Glucose

-0.09

-0.124 = BMI

+0.04

-0.446 = Age

+0.02

-0.695 = SkinThickness

-0.01

-0.363 = BloodPressure

-0

-0.251 = Pregnancies

-0

0.20

0.25

0.30

0.40

$E[f(X)] = 0.346$

